

1.1 Gradient Descent

- Error: $\frac{1}{2}(t-o)^2 \leftarrow$ Back propagation for a single neuron unit

- Gradient Descent for output unit

$$\Delta w_i = -\eta \frac{\partial E}{\partial w_i}, \quad \frac{\partial E}{\partial w_i} = \frac{\partial E}{\partial o} \cdot \frac{\partial o}{\partial w_i} \rightarrow \frac{\partial E}{\partial o} = \frac{\partial}{\partial o} \left[\frac{1}{2}(t-o)^2 \right] = -(t-o) \frac{\partial}{\partial o}$$

$\uparrow$ Find bias weight and following weights

$$\bullet \text{ Given } o = w_0 + \sum_{i=1}^n w_i(x_i + x_i^2)$$

$$\bullet \frac{\partial o}{\partial w_0} = 1 \quad ; \quad \frac{\partial E}{\partial o} = -(t-o) \cdot 1 \leftarrow \text{Bias weight}$$

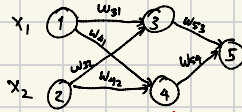
$$\bullet \frac{\partial o}{\partial w_i} = (x_i + x_i^2) \quad ; \quad \frac{\partial E}{\partial w_i} = -(t-o)(x_i + x_i^2)$$

- Final training rule
 $\Delta w_i = \eta(t-o)(x_i + x_i^2)$ for
 $i = 1, 2, \dots, n$

1.2 Comparing Activation Function

a) Node 3 output: $h(w_{31}x_1 + w_{32}x_2)$

Node 4 output: $h(w_{41}x_1 + w_{42}x_2)$



$$\text{Node 5 output } (y_5): h(w_{53}(h(w_{31}x_1 + w_{32}x_2)) + w_{54}(h(w_{41}x_1 + w_{42}x_2)))$$

b) Given $X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$, $W^{(1)} = \begin{pmatrix} w_{3,1} & w_{3,2} \\ w_{4,1} & w_{4,2} \end{pmatrix}$, $W^{(2)} = \begin{pmatrix} w_{5,3} & w_{5,4} \end{pmatrix}$

(Node 3) output: $h(W^{(1)}X) = h \begin{pmatrix} w_{3,1}x_1 + w_{3,2}x_2 \\ w_{4,1}x_1 + w_{4,2}x_2 \end{pmatrix}$

$$W^{(1)}X = \begin{pmatrix} w_{3,1}x_1 + w_{3,2}x_2 \\ w_{4,1}x_1 + w_{4,2}x_2 \end{pmatrix}$$

$$W^{(2)}(h(W^{(1)}X)) = \begin{pmatrix} w_{5,3}(w_{3,1}x_1 + w_{3,2}x_2) \\ w_{5,4}(w_{4,1}x_1 + w_{4,2}x_2) \end{pmatrix}$$

$$y_5: h(W^{(2)}h(W^{(1)}X)) = h \begin{pmatrix} w_{5,3}(w_{3,1}x_1 + w_{3,2}x_2) \\ w_{5,4}(w_{4,1}x_1 + w_{4,2}x_2) \end{pmatrix}$$

c) Given $h_s(x) = \frac{1}{1+e^{-x}}$ and $h_t(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

$$\hookrightarrow 2 \cdot h_s(2x) = \frac{2}{1+e^{-2x}} \rightarrow 2 \cdot h_s(2x) - 1 = \frac{2}{1+e^{-2x}} - \frac{1+e^{-2x}}{1+e^{-2x}}$$

$$\hookrightarrow \frac{1-e^{-2x}}{1+e^{-2x}} \rightarrow \frac{1-e^{-2x}}{1+e^{-2x}} \cdot \frac{e^x}{e^x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$h_t(x) = 2 \cdot h_s(2x) - 1$$

Ex: $y_5 = h_t(w_{5,3}h_t(w_{3,1}x_1 + w_{3,2}x_2) + w_{5,4}h_t(w_{4,1}x_1 + w_{4,2}x_2)) \leftarrow$ using $h_t(x)$

sigmoid conversion $\left(w_{5,3}(2h_s(2w_{3,1}x_1 + 2w_{3,2}x_2) - 1) + w_{5,4}(2h_s(2w_{4,1}x_1 + 2w_{4,2}x_2) - 1) \right) \leftarrow$ inside of $h_t(x)$ of y_5

$$y_5 = 2h_s \left[2(w_{5,3}(2h_s(2w_{3,1}x_1 + 2w_{3,2}x_2) - 1) + w_{5,4}(2h_s(2w_{4,1}x_1 + 2w_{4,2}x_2) - 1)) \right] = 1 \leftarrow \text{using } h_s(x)$$

- Because tanh can be converted to sigmoid using linear transformed weights and constant, they can generate the same function.